

Agent Based Modeling

Rock, Paper, Scissors

KAYLEN TON

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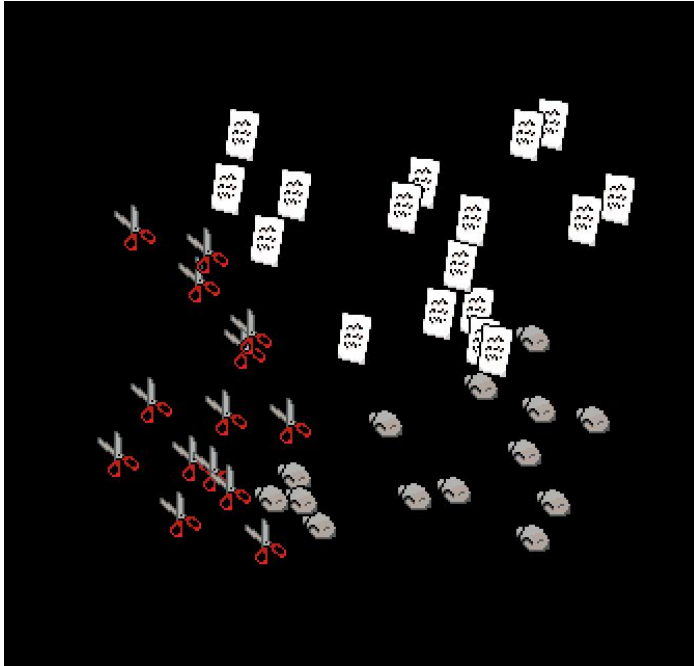
The End

Open to questions.

01

Introduction

Inspirations

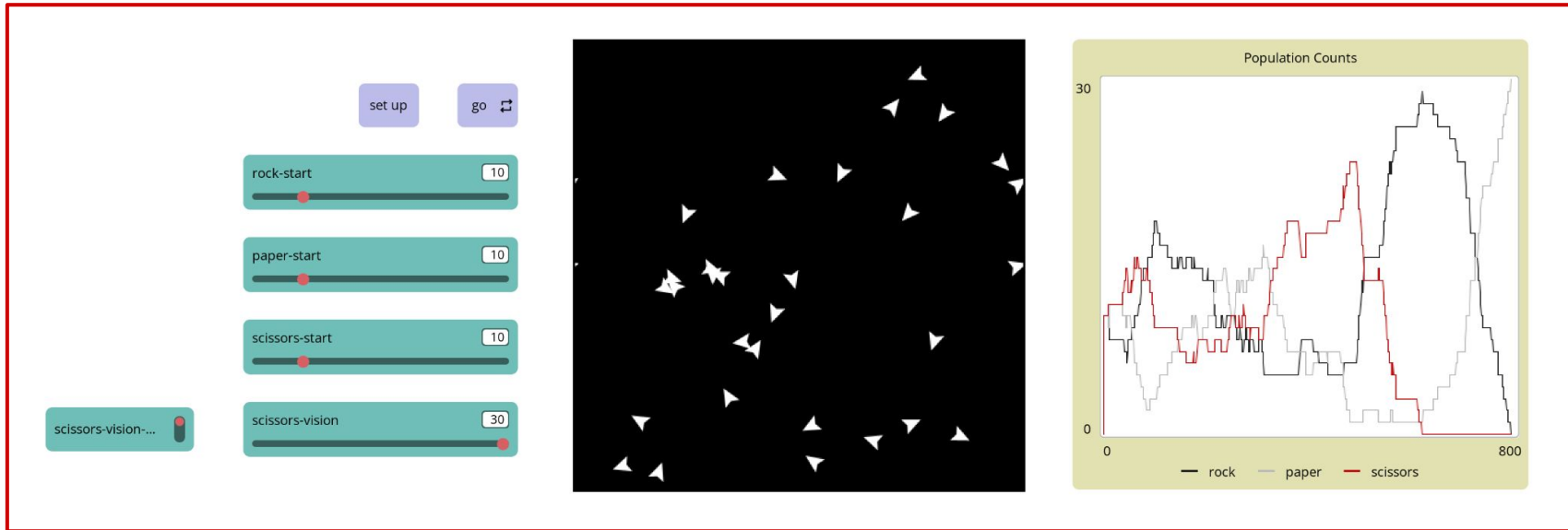


& The Sheep Predator Prey ABM

- From the NetLogo Model Library

Wilensky, U. (1997). NetLogo Wolf Sheep Predation model.
<http://ccl.northwestern.edu/netlogo/models/WolfSheepPredation>. Center for Connected Learning and Computer-Based Modeling, Northwestern University, Evanston, IL.

- I like when a model's outcome can be flipped in seconds.
- Simplicity
- Simple summary and aggregation functions (COUNT)



A very simple model.

02

Model Components

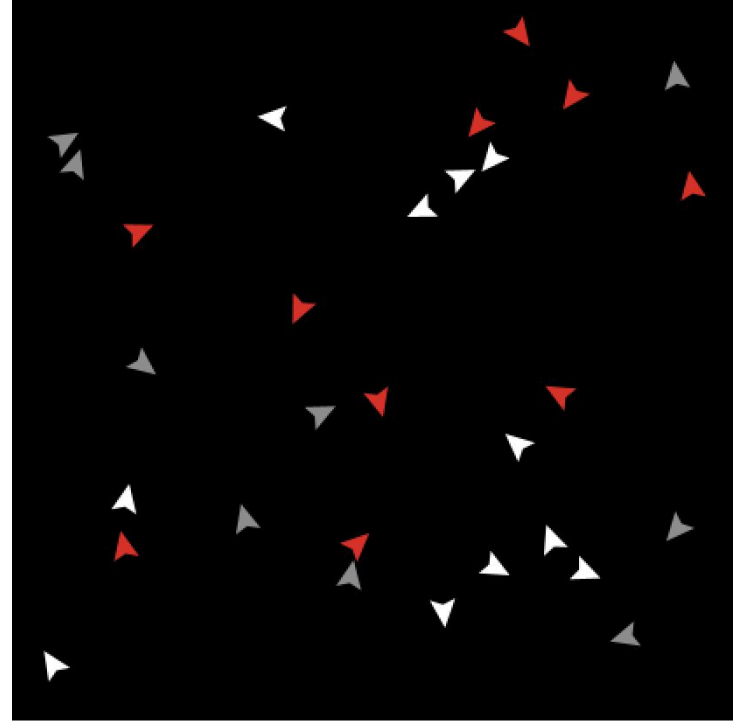
Agent Breeds, Environment

Three turtle agent breeds:

- Rock (Gray)
- Paper (White)
- Scissors (Red)

World settings:

- Center
- min-pxcor and pycor = -20
- max-pxcor and pycor = 20
- Wrapped both horizontally and vertically



Buttons, Sliders, and Switch

Change the starting population for each turtle breed.

- Min: 1
- Max: 50

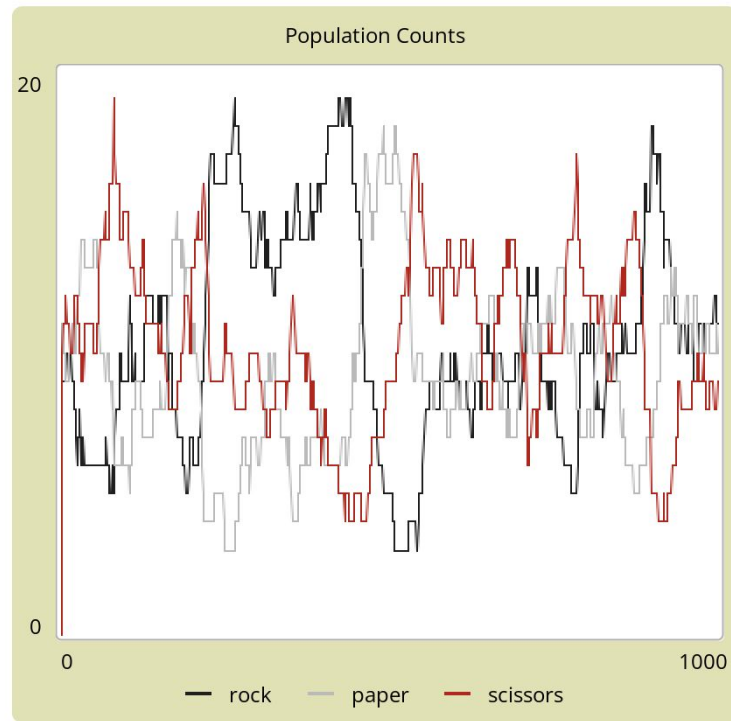
Enable a vision radius for the *scissors* team.

- Min: 0
- Increment: 5
- Max: 30

The image shows a control panel for a NetLogo simulation, enclosed in a red rectangular border. At the top, there are two buttons: a light purple button labeled "set up" and a black button labeled "go" with a circular arrow icon. Below these are four sliders, each with a teal background and a dark grey track. The first three sliders are labeled "rock-start", "paper-start", and "scissors-start", each with a value of 10 displayed in a white box on the right. The fourth slider is labeled "scissors-vision" with a value of 0. To the left of the "scissors-vision" slider is a switch control labeled "scissors-vision-..." with a red indicator light.

Plot Widget

Per tick, this graph tracks the counts for each turtle breed population.



Types of Procedures

Movement

- By default, every breed spawns randomly on the world and just goes forward one.
- Scissors-vision-on? = ON
 - If the agent sees a prey, it will chase.
 - If it sees a predator, it will run/avoid.

Interaction

- When different turtle breeds collide/cross paths, the loser becomes the winner.
 - Rock & scissors → scissors becomes rock.
 - Scissors & paper → paper becomes scissors.

Types of Procedures (cont.)

Tracking / Plotting

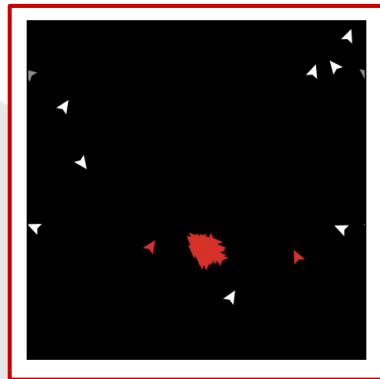
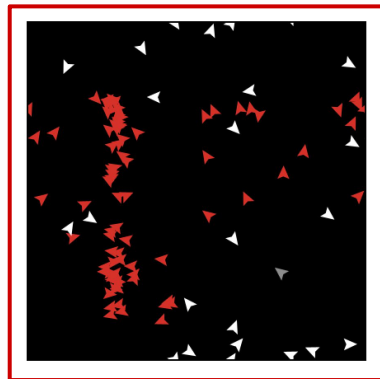
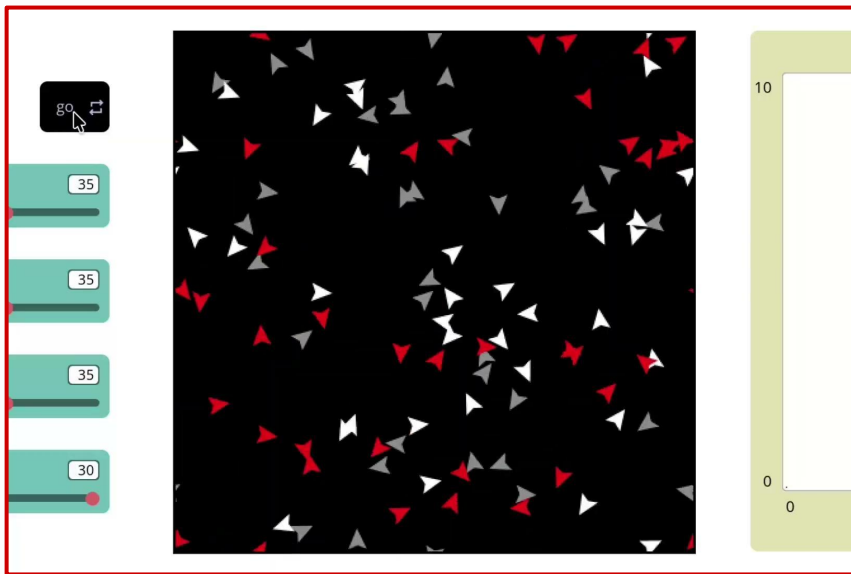
```
98  to become-rock
99      set breed-type "rock"
100      set color gray
101  end
102
103  to become-paper
104      set breed-type "paper"
105      set color white
106  end
107
108  to become-scissors
109      set breed-type "scissors"
110      set color red
111  end
112
113  to update-counts
114      set rock-count count turtles with [breed-type = "rock"]
115      set paper-count count turtles with [breed-type = "paper"]
116      set scissors-count count turtles with [breed-type = "scissors"]
117  end
```

Misc.

- After 1000 ticks, if there is no winner (where one breed takes over the entire world population of turtle agents), the simulation stops.

03

Running the Model



04

My Experiment

RQs.

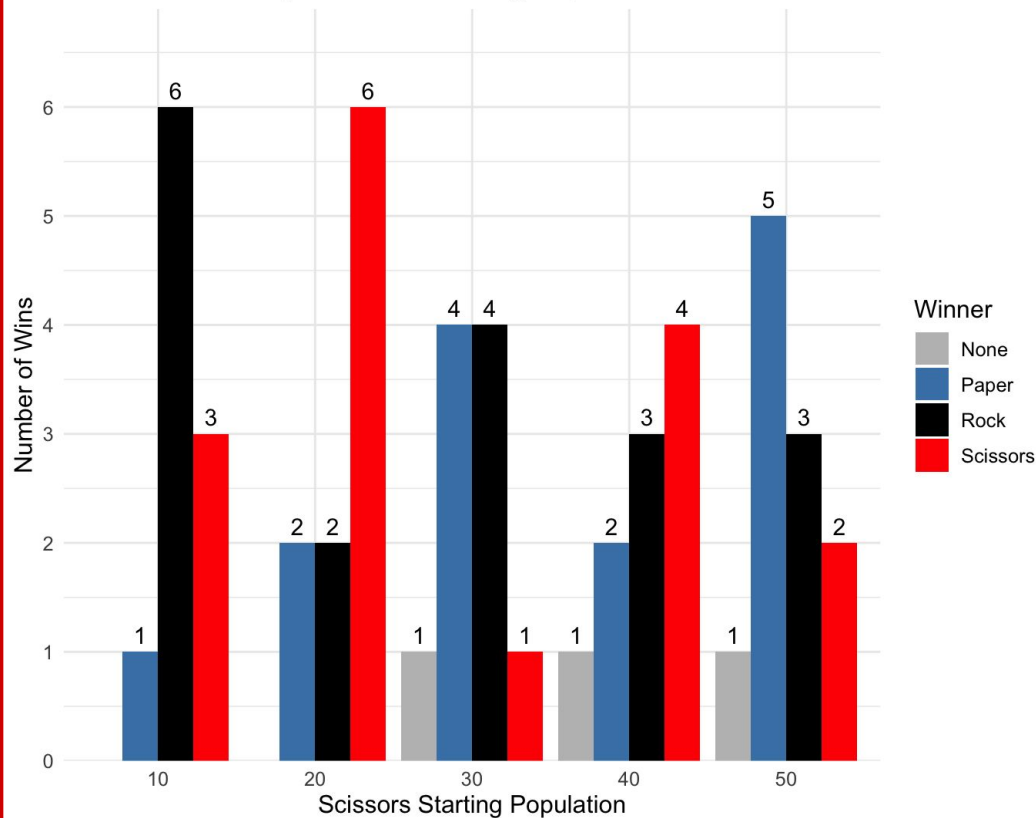


How do different starting populations for scissors affect the outcome of which agent comes out on top?



How do different vision radius ranges affect the outcome of which agent comes out on top?

Win Outcomes by Scissors Starting Population

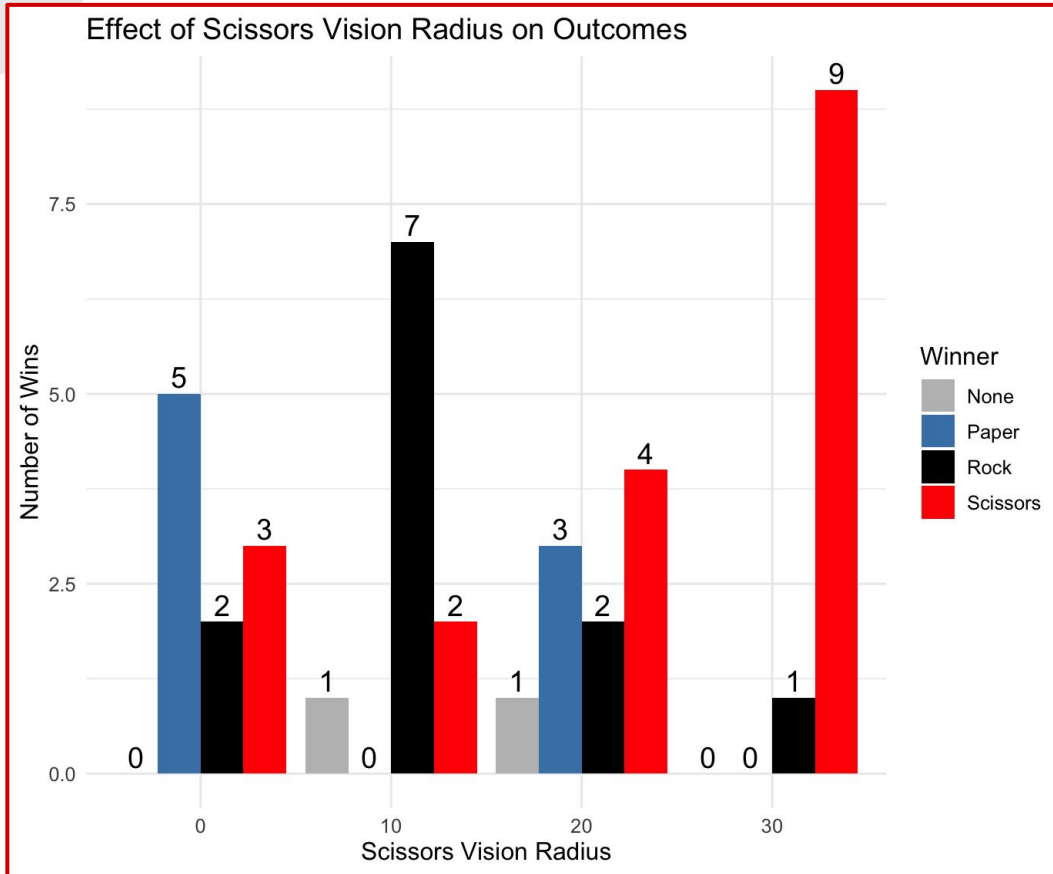


1. Starting Pop.

10 Simulations, changing the initial population of scissors from 10 to 50 in increments of 10.

Rock and paper both started at 20 for each simulation.

Scissors-vision-on? = OFF



2. Vision Radius

10 Simulations each, changing team scissors' vision radius by increments of 10 from 0 to 30.

All starting populations are 30.

Scissors-vision-on? = ON

05

Conclusion

RQ 1 - Starting Pops. Results

Low scissors → Rock dominates

Mid range → mixed / unstable

High scissors → Paper starts winning more

Increasing scissors does **NOT** guarantee scissors dominance. Instead, it can create a **delayed advantage** for its predator breed.

RQ 2 - Vision Radius

0 vision → Paper wins 5/10 (baseline randomness)

10 vision → Rock dominates 7/10 (killed all paper, left with rocks)

20 vision → Scissors wins 4/10. More balanced. (Rock 2; Paper = 3)

30 vision → Scissors dominates heavily, winning 9/10.

Having a vision radius at higher level differences (rock and paper have zero) gives a greater chance at winning. ***Vision radiuses are advantageous.***

If I could redo this...

- More simulations
- Wider parameter ranges
- More granular increments when changing parameter ranges

06

The End